

# Business Use Case: Eleven RAG Chatbot for Customer Support Automation

## Project Title

**Eleven RAG Chatbot – AI-Powered Customer Support Assistant using Retrieval-Augmented Generation (RAG)**

## Business Problem

E-commerce businesses often receive a large number of repetitive customer support queries related to product availability, return policies, delivery timelines, payment options, and company policies. Handling these requests manually increases support costs, creates longer response times, and leads to unnecessary support ticket generation.

As the customer base grows, hiring additional support agents becomes expensive and negatively impacts operational efficiency.

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## Objective

Develop an AI-powered customer support chatbot that answers customer queries using the company's inventory records and policy documents, thereby reducing support workload, improving customer experience, and lowering operational costs.

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## Solution

Built a **Retrieval-Augmented Generation (RAG) chatbot** using **LM Studio** with a locally hosted Large Language Model (LLM).

Instead of relying only on the model's pre-trained knowledge, the chatbot retrieves relevant information from:

- Product inventory database
- Shipping policies
- Return & refund policies
- Payment policies
- Frequently Asked Questions (FAQs)
- Internal company knowledge documents

The retrieved context is combined with the LLM to generate accurate, context-aware responses.

This approach minimizes hallucinations while ensuring answers remain aligned with company-specific information.

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## Business Workflow

1. Customer asks a question through the chatbot.
  2. The query is converted into vector embeddings.
  3. The RAG engine searches the knowledge base for the most relevant inventory records or policy documents.
  4. Retrieved information is passed to the locally hosted LLM in LM Studio.
  5. The LLM generates a contextual and accurate response.
  6. If the chatbot cannot confidently answer the query, it escalates the conversation to a human support agent.
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## Key Features

- AI-powered natural language conversation
  - Company-specific knowledge retrieval
  - Inventory-aware responses
  - Policy-based question answering
  - Reduced hallucination through RAG architecture
  - Offline inference using LM Studio
  - Human-agent escalation for complex cases
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## Business Value Delivered

### 1. Reduced Customer Support Cost

The chatbot automatically resolves repetitive customer queries, reducing the number of support tickets that require human intervention.

### 2. Lower Operational Cost

Fewer support tickets mean fewer customer support resources are needed for routine questions, allowing teams to focus on higher-value issues.

### 3. Faster Response Time

Customers receive instant responses instead of waiting for support agents, improving overall customer satisfaction.

### 4. Improved Customer Experience

Customers receive consistent, accurate, and policy-compliant answers 24/7.

## 5. Better Agent Productivity

Support agents spend less time answering repetitive questions and more time resolving complex customer issues.

## 6. Consistent Information Delivery

Since responses are grounded in official company documents, customers receive standardized information across all interactions.

## 7. Scalable Customer Support

The chatbot can handle thousands of simultaneous customer conversations without increasing support headcount.

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# Product Metrics (KPIs)

## North Star Metric

**Customer Queries Resolved Without Human Intervention (%)**

## Success Metrics

- Support Ticket Reduction (%)
- First Response Time (FRT)
- Average Resolution Time (ART)
- Self-Service Resolution Rate
- Customer Satisfaction Score (CSAT)
- Customer Effort Score (CES)
- Escalation Rate
- Average Handling Time (AHT)
- Monthly Customer Support Cost
- Knowledge Retrieval Accuracy

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# Technical Architecture

- **LLM:** Local model hosted using LM Studio
- **AI Technique:** Retrieval-Augmented Generation (RAG)
- **Knowledge Base:** Inventory records, company policies, FAQs
- **Vector Search:** Semantic similarity search using embeddings
- **Response Generation:** Retrieved context + LLM prompt
- **Fallback Mechanism:** Escalation to human support when confidence is low

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## Expected Business Impact

- 40–60% reduction in repetitive customer support tickets
- 24/7 availability without increasing support staffing
- Faster customer response times
- Improved customer satisfaction through instant assistance
- Reduced operational expenditure on customer support
- Better utilization of customer support agents for complex issues
- Higher consistency and accuracy in customer communication

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## Product Manager Perspective

From a Product Management standpoint, this solution addresses a clear business pain point by automating repetitive support interactions while maintaining response accuracy through Retrieval-Augmented Generation. Instead of replacing customer support agents, the chatbot augments the support process by handling low-complexity, high-frequency queries and seamlessly escalating complex cases to humans. This improves operational efficiency, reduces support costs, enhances customer experience, and creates a scalable customer support system that grows with the business.